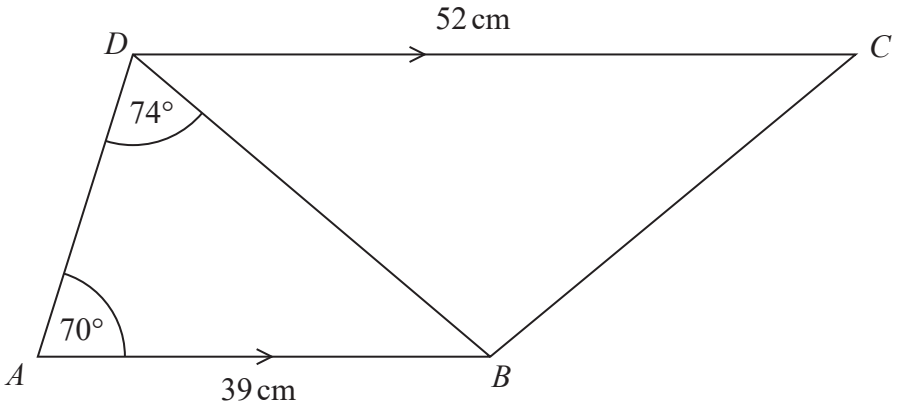
15 Here is trapezium *ABCD*.



AB and *DC* are parallel.

Work out the area of triangle *BCD*.
Give your answer correct to 3 significant figures.

In triangle ABD, use the sine rule:

1 mark
Sine rule in ABD.

$$\frac{DB}{\sin 70°} = \frac{39}{\sin 74°}$$

$DB = 39\sin 70° \div \sin 74° = 38.124906...$ cm

1 mark
Find DB.

Angle ABD = 180° − 70° − 74° = 36°

1 mark
Find the included angle.

Angle BDC = 36° (alternate angles, AB parallel to DC)

Area BCD = ½ × DC × DB × sin(angle BDC)

1 mark
Use ½ab sinC.

$= ½ \times 52 \times 38.124906... \times \sin 36°$

$= 582.640701...$ cm^2 = 583 cm^2 (3 s.f.)

1 mark
583 cm^2.

..... cm^2

(Total for Question 15 is 5 marks)